Nonexistence of $(243, 121, 60)$ difference sets in $C_3 \times C_9 \times C_9$

P. Graham
https://orcid.org/0009-0007-8493-1505

October 21, 2026

Abstract

A (v, k, λ) difference set in a finite group of order v is a k-element subset in which every nonidentity element occurs exactly λ times as an ordered difference. We prove, by exhaustive computation, that no $(243, 121, 60)$ difference set exists in $C_3 \times C_9 \times C_9$. This group-specific case appears as unresolved in the 1997 table of Vera López and García Sánchez and was singled out by Gordon in 2019 as the "Smallest Open Case." A multiplier-fixed translate has thirteen points in the subgroup of elements annihilated by three. Moment identities, singleton congruences, and legal affine normalizations reduce the problem to eight quotient multiplicity arrays, each with a complete ternary lifting tree. Integer difference-count searches exhaust these trees, with separate checks of decomposition completeness. Lean proves the multiplier reduction, normalization, lifting coverage, and pruning lemmas; its final nonexistence theorem is explicitly conditional on the externally established search assertions.

The proofs, the Lean formalization, the search code, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 The result and its historical setting

Write $C_m = \mathbb{Z}/m\mathbb{Z}$ additively. For a subset P of an abelian group, put

$$d_P(g) = \big|\{(a, b) \in P \times P : a - b = g\}\big|.$$

A (v, k, λ) difference set in a group of order v is a subset D with $|D| = k$ and $d_D(g) = \lambda$ for every $g \neq 0$.

Theorem 1. *There is no subset $D \subseteq C_3 \times C_9 \times C_9$ such that $|D| = 121$ and*

$$\big|\{(a, b) \in D \times D : a - b = g\}\big| = 60 \qquad \textit{for every } g \neq 0.$$

The theorem is computer-assisted. Its scope is exactly these parameters in this group. The mathematical reduction and its coverage are proved in Lean; the exhaustive integer computations that reject the covered leaves are external to Lean. Section 10 states this boundary precisely.

Vera López and García Sánchez undertook a systematic study of abelian difference sets with $100 < k \leq 150$ [14]. Their Table 1, printed page 105, marks the present group and parameters with a question mark, meaning unknown according to their convention on page 101. That paper was published in 1997; the publisher gives February 28, 1997. This supplies a published unresolved-status record, without asserting that it was the first occasion on which the question was asked.

Gordon's La Jolla Difference Set Repository has made such group-specific questions accessible and maintained their connection with the literature. In his presentation of August 3, 2019, Gordon singled out the present case as the "Smallest Open Case" [3, slide 38]. Here slide 38 is physical PDF page 49. The repository's archived version 1.3, dated April 24, 2026, also records the exact entry $\mathrm{DS}(243, 121, 60, [3, 9, 9])$ as Open [4]. The entry has a status and an empty comment, with no per-entry date. The archive date therefore does not date the original insertion or last review of this entry. Similarly, the website's September 3, 2026 update label is a site-level date. The entry's original insertion and last-review dates have not been recovered.

These sources also put the question in perspective. For the same parameters, the other six abelian groups of order 243 are already recorded as follows; this is historical context, not a new proof of those entries:

group	recorded answer	attribution
C_{243}	No	Lander
$C_3 \times C_{81}$	No	Lander
$C_9 \times C_{27}$	No	Arasu–Ma
$C_3^2 \times C_{27}$	No	Vera López–García Sánchez
$C_3^3 \times C_9$	No	Vera López–García Sánchez
C_3^5	Yes	Paley's finite-field construction

The source for the complete list and its attributions is Gordon [4].[1] The 1997 table independently records the negative entries for C_{243}, $C_3 \times C_{81}$, $C_3^2 \times C_{27}$, and $C_3^3 \times C_9$; its method 21 uses multiplier-orbit enumeration with multiplier 61. Arasu and Ma [1] subsequently resolved $C_9 \times C_{27}$. For C_3^5, Paley's quadratic-residue construction [12] in the additive group of $\mathbb{F}_{243}$ gives the stated parameters. The 1997 table cites Emma Lehmer [8] for its positive entry. The nonexistence at issue here concerns the remaining group $C_3 \times C_9 \times C_9$.

The work uses the classical multiplier and character methods developed by Hall [5], Menon, Turyn, and others, and the systematic multiplier searches of Vera López and García Sánchez. It also benefits from the later nonexistence and field-descent theory of Arasu, Ma, Leung, Schmidt, and Miller. These contributions supply both the historical question and the algebraic framework for reducing it to a finite computation. The contributors to Lean, mathlib, and the declared library dependencies are also acknowledged for the formal-development infrastructure.

2 Published restrictions and their hypotheses

Set

$$v = 243 = 3^5, \qquad k = 121, \qquad \lambda = 60, \qquad n = k - \lambda = 61, \qquad \exp(G) = 9.$$

The parameter identity $k(k-1) = \lambda(v-1)$ holds. The odd-order Bruck–Ryser–Chowla equation $61x^2 - 60y^2 = z^2$ has the nonzero solution $x = y = z = 1$. These elementary necessary conditions do not exclude the parameters.

Turyn's character restrictions [13] allow this application: $\mathrm{ord}_3(61) = 1$ is odd. At the nontrivial character orders 3 and 9, the prime 61 is not self-conjugate. Indeed, its powers modulo 3 equal 1, and modulo 9 cycle through $7, 4, 1$, never -1. Obstructions requiring a common prime divisor of n and v have no such prime here; in particular that ingredient of the Lander-type restrictions in

[1] This small status summary is selected and reformatted from Daniel M. Gordon's repository data, copyright 2024 Daniel M. Gordon, under CC BY 4.0. Group notation and presentation are changed; the source status values are unchanged. No endorsement is implied.

Leung–Ma–Schmidt [9] is absent. That comparison uses the stated common-prime setting, without claiming a full reading of that paper's proof. The multiplier theorem does apply and provides symmetry, as proved in Section 3; that symmetry alone is not a nonexistence result.

For the field-descent function of Leung and Schmidt [10, Definition 2.6], one obtains $F(9,61) = 3$. Here $\mathrm{ord}_3(61) = 1$ and $61 \not\equiv 1 \pmod 9$, so the exponent $b_3 = 1$ is permitted in that definition. Taking their factor A to be the entire group and the complementary factor trivial, Theorem 4.3 gives

$$\exp(G) \le 243\sqrt{\frac{3^2}{4 \cdot 61 \cdot \varphi(3)}} = \frac{729}{\sqrt{488}} \approx 33.0003,$$

which permits exponent 9. Their Theorem A cannot instead split off a C_9 factor: its coprimality condition would require $\gcd(9, \exp(C_3 \times C_9)) = 1$, whereas this gcd is 9. Theorem 4.1 requires a cyclic Sylow subgroup, also absent here.

Arasu and Ma's 2001 nonexistence result [1] resolves $C_9 \times C_{27}$. The relevant theorem, as explicitly restated and extended by Miller [11, Theorems 1 and 3], assumes

$$G = C_{p^e} \times J, \qquad e \ge 2, \qquad \exp(\mathrm{Syl}_p J) < p^e.$$

Splitting off C_9 in the present group leaves another C_9, so the strict exponent inequality fails. The forbidden-multiplier congruences would demand both $t \equiv 4 \pmod 9$ and $t \equiv 1 \pmod 9$. Although the older numerical condition $4(p-1)n = 488 > 243$ holds, it cannot repair the failed group hypothesis. This comparison uses Miller's restatement; the full 2001 original was not available for a complete reading.

Chen, Xiang, and Sehgal [2] study skew Hadamard abelian difference sets, which satisfy the additional partition $G = \{0\} \dot\cup D \dot\cup (-D)$. The present parameters do not supply this hypothesis. Consequently a skew-only exponent bound cannot be applied to the ordinary difference-set question. This scope distinction follows from the published abstract; the full original was not obtained, and no numerical specialization of its bound is needed here.

Recent work on signed difference sets by He and Wu [6] includes ordinary difference sets but does not yield this exclusion by the criteria considered here. Their Theorem 3.2 has the same failed self-conjugacy test for 61; Theorem 4.2 requires even v; Theorem 4.4 requires $k = v - 1$; and Theorem 4.7 requires an elementary abelian group and a defect bound, whereas this group is not elementary abelian and the defect is $d = 122 > 5$. Miller's and He–Wu's cited versions are preprints, not claimed here as verified journal publications.

These are explicit applications and failures of hypotheses, not a claim to have exhausted all uses of these methods. The historical Open entries document the question; they cannot establish an exhaustive priority claim. Some older originals, including Lander's Theorem 4.38 [7] and the Hall, Paley, and Lehmer works cited for historical credit, were not read in full. Hall's survey is not used to infer that a cyclic-group statement alone covers every abelian hypothesis of the multiplier specialization. Theorem 1 rests on the reduction and exhaustion given here, rather than on an inference from an Open label or from the absence of a located earlier answer.

3 Multiplier fixedness and quotient moments

Use coordinates $G = C_3 \times C_9 \times C_9$ and put

$$H = G[3] = \{(x, 3u, 3w) : x, u, w \in C_3\},$$
$$K = 3G = \{(0, 3u, 3w) : u, w \in C_3\}.$$

Thus $|H| = 27$, $|K| = 9$, and $Q = G/K \cong C_3^3$. In particular, H and K are different subgroups. Elements of G are encoded by $81x + 9y + z$, using the least nonnegative coordinate representatives; elements of Q by $9x + 3\bar{y} + \bar{z}$.

Lemma 1. *If D is a difference set as in Theorem 1, then a translate D_0 satisfies $61D_0 = D_0$.*

Proof. The first multiplier theorem applies because G is abelian, $n = 61$ is prime, $61 > \lambda = 60$, and $\gcd(61, 243) = 1$. It gives $61D = D + g$ for some $g \in G$. This is also the $n_1 = 61$ specialization of Menon's multiplier theorem in Miller [11, Theorem 5]; see the classical character and multiplier theory in Turyn [13]. Put $s = \sum_{d \in D} d$. Summing the set identity gives $60s = 121g$. Multiplication by 121 has inverse multiplication by 7 on G, since $121 \cdot 7 \equiv 1 \pmod 9$. For $u = 7s$ we therefore have $60u = g$, whence

$$61(D - u) = D + g - 61u = D - u.$$

Take $D_0 = D - u$. □

The initial translating element u can be anywhere in G. Subsequent normalizing translations will lie in H. Multiplication by 61, equivalently by 7 modulo 9, has order three and fixed subgroup exactly H. Its scalar orbits consist of 27 singletons and 72 triples. Every fixed subset is a union of these orbits. A group automorphism commutes with scalar multiplication, and translation by $t \in H$ preserves fixedness because $60t = 0$.

A difference set has no nonzero translation period: if $D + h = D$, then $d_D(h) = 121$, contrary to 60 when $h \neq 0$. It follows that the fixed translates of a fixed difference set D_0 are exactly $D_0 + H$: fixedness of $D_0 + t$ implies that $60t$ is a period, hence $60t = 0$, whose solution set in G is H.

For later use, let $U \leq G$ and $r_U(a+U) = |D \cap (a+U)|$. Counting ordered pairs whose difference lies in U gives

$$\sum_{a+U} r_U(a + U) = 121, \qquad \sum_{a+U} r_U(a + U)^2 = 121 + 60(|U| - 1). \tag{1}$$

More generally, projected autocorrelation equals $121 + 60(|U| - 1)$ at zero and $60|U|$ elsewhere. For an index-three subgroup, the three coset counts consequently satisfy

$$t_0 + t_1 + t_2 = 121, \qquad t_0^2 + t_1^2 + t_2^2 = 4921, \qquad 0 \leq t_i \leq 81. \tag{2}$$

The complete integer solutions are the six permutations of $(36, 40, 45)$. This small finite assertion is checked both in the finite enumerations and in Lean.

4 The thirteen singleton points

Assume from now on that D_0 is fixed, and put $S = D_0 \cap H$. There are four projective nonzero linear forms on $(\bar{y}, \bar{z}) \in C_3^2$. For each, the two nonzero fibers contain only scalar triples, so their intersection counts with D_0 are divisible by three. Of $36, 40, 45$, only 40 is not divisible by three; thus the zero-fiber count is 40. Every point of H lies in all four zero fibers, and every point outside H lies in exactly one. Hence

$$4 \cdot 40 = 4|S| + (121 - |S|), \qquad |S| = 13.$$

Let a, b, c be the sizes of S in the three x-layers of H. They satisfy $a + b + c = 13$ and $0 \leq a, b, c \leq 9$. On $Q = G/K$, the three special cells $(x, 0, 0)$ contain only singleton points and

have multiplicities a, b, c. Every other cell contains three scalar triples, so its multiplicity lies in $\{0, 3, 6, 9\}$. The $U = K$ instance of (1) gives

$$a^2 + b^2 + c^2 \equiv 601 \equiv 7 \pmod 9.$$

The x-quotient in (2) forces the residues of (a, b, c) modulo three to be a permutation of $(1, 0, 0)$. The complete list of layer triples satisfying these conditions consists of the permutations of $(4, 0, 9)$ and $(4, 3, 6)$.

For any $h \in H \setminus \{0\}$, the scalar action permutes the 60 ordered pairs in D_0^2 of difference h. Its fixed pairs have both endpoints in S; all remaining pair orbits have length three. Therefore

$$|S \cap (S + h)| \equiv 0 \pmod 3 \qquad (h \in H \setminus \{0\}). \tag{3}$$

A $(4, 0, 9)$ layer pattern has intersection four with its translate by $(1, 0, 0)$, so cannot satisfy (3).

For an explicit finite encoding, the point $(x, 3u, 3w) \in H$ has mask bit $9x + 3u + w$, with bit zero the least significant bit. The layer conditions above describe exactly

$$6\left[\binom{9}{4} + \binom{9}{4}\binom{9}{3}\binom{9}{6}\right] = 5{,}335{,}092 \tag{4}$$

distinct subsets of H. This is a count of candidate singleton subsets, not of difference sets or of all fixed 121-point sets. Exhaustive testing of all 26 congruences leaves exactly 7,776 masks. An enumeration scanning every one of the $\binom{27}{13} = 20{,}058{,}300$ thirteen-point subsets independently reproduces both counts.

The surviving set of masks is exactly the orbit of

$$S_0: \quad \text{mask } 24909666, \qquad (a, b, c) = (4, 3, 6), \tag{5}$$

under the legal affine maps described next. Equality of these two finite sets, with normalization witnesses, establishes normalization; agreement of their cardinalities alone would not do so. Thus every hypothetical difference set has a fixed legal image with H-part exactly S_0.

5 Legality of the normalizing symmetries

Use $F = A + t$ with $t = (t_0, 3t_1, 3t_2) \in H$, where

$$\begin{aligned} F(x, y, z) = (&ax + by + cz + t_0 \pmod 3, \\ &3dx + ey + fz + 3t_1 \pmod 9, \\ &3gx + hy + iz + 3t_2 \pmod 9). \end{aligned} \tag{6}$$

Here $a \not\equiv 0 \pmod 3$ and $B = \left(\begin{smallmatrix} e & f \\ h & i \end{smallmatrix}\right)$ is invertible modulo three. The parameters in the first coordinate and d, g, t_j are taken modulo three; entries of B have chosen lifts modulo nine.

The coordinate orders make A additive. Its induced map on $G/3G$ has block upper-triangular matrix with diagonal blocks a and $B \pmod 3$, so is invertible. If $A(p) = 0$, it follows that $p \in 3G = K$. On K, the map is again given by $B \pmod 3$, which is invertible; hence $p = 0$. Thus A is an automorphism. Translation preserves differences, and $60t = 0$ gives $F(61p) = 61F(p)$. Consequently F preserves the difference-set property and scalar fixedness. The affine map F need not fix zero; it is its linear part A that is a group automorphism.

For $b = c = 0$, the induced H-actions, writing its last two coordinates as a vector $\boldsymbol{u} \in C_3^2$, are

$$(x, \boldsymbol{u}) \longmapsto (ax + t_0, B\boldsymbol{u} + \boldsymbol{d}x + \boldsymbol{t}).$$

They form a group of order $2{\cdot}3{\cdot}48{\cdot}9{\cdot}9 = 23{,}328$. Each has an explicit full-G lift of (6). All these lifts have been checked on the full 243-point group for bijectivity, translation in H, scalar commutation, and additivity of the linear part on every generator edge. The generator-edge identities imply additivity for arbitrary pairs by telescoping along a coordinate-generator path. The orbit of S_0 has size 7,776, so its stabilizer in this H-action group has order three.

Since the three layer sizes of S_0 are distinct, an S_0-stabilizing H-action has $a = 1, t_0 = 0$. Each of its three possibilities has nine (b, c) extensions. These induce 27 distinct permutations on Q, forming a group of order 27. Their chosen full-G lifts preserve S_0, commute with the scalar, and act consistently on every K-coset. These facts are checked for the actual point maps, including full pairwise additivity in the separate quotient-map check. The chosen lifts need not themselves form a 27-element subgroup of affine maps on G. The stabilizer of order three on H and the induced group of order 27 on Q refer to different actions; neither is asserted to be the full affine stabilizer of S_0 on G.

6 Complete quotient multiplicity classification

Normalize S to S_0 and define

$$r(q) = |D_0 \cap (q + K)| \qquad (q \in Q).$$

The notation now refers to the normalized fixed image of D_0. This is an integer multiplicity array on G/K, not a subset of G and not an array on G/H. Its special entries $r(x, 0, 0)$ are $(4, 3, 6)$; initially each of the other 24 entries may be any of $0, 3, 6, 9$.

Choose one representative of each of the 13 projective nonzero forms $L : Q \to C_3$, by requiring its first nonzero coefficient to be 1, and put

$$t_L(j) = \sum_{L(q)=j} r(q).$$

If L ignores x, its zero fiber contains all thirteen singleton points; otherwise it contains the four points of the $x = 0$ singleton layer. Thus $t_L(0) \equiv 1 \pmod 3$ in both cases. Equation (2) forces

$$t_L(0) = 40, \qquad \{t_L(1), t_L(2)\} = \{36, 45\}. \tag{7}$$

For $p, q \in Q$, exactly thirteen of the forms have $L(p) = L(q)$ when $p = q$, and exactly four when $p \neq q$. Therefore

$$\sum_L t_L(L(q)) = 13r(q) + 4(121 - r(q)) = 484 + 9r(q). \tag{8}$$

This identity proves completeness of the following finite procedure. Enumerate all $2^{13} = 8192$ ways of ordering $(36, 45)$ in (7); recover $r(q)$ from (8); retain integer coefficients between zero and nine, the special entries $(4, 3, 6)$, and nonspecial entries divisible by three. Every possible quotient array must occur, since its line assignments occur and the inversion recovers it uniquely.

Exactly 144 arrays survive. Each has projected autocorrelation 601 at zero and 540 at every nonzero element of Q, as required by (1) and its nonzero counterpart. Each has twelve nonspecial entries equal to three and twelve equal to six. The absence of zero and nine is an enumeration

result, not an assumption used to exclude possible arrays. A separate depth-first enumeration assigns $\{0, 3, 6, 9\}$ to the 24 nonspecial cells and enforces all index-three moment triples, without assuming which fiber has count 40. It gives exactly the same set of 144 arrays.

The group of 27 induced S_0-stabilizing quotient permutations partitions these arrays into eight orbits. Representatives in the order used for lifting are given in Table 1. Each nine-digit string lists $r(x, \bar{y}, \bar{z})$ for fixed x, with $3\bar{y} + \bar{z} = 0, \ldots, 8$ in increasing order. Their orbits are disjoint and their

case c	orbit size	$x = 0$	$x = 1$	$x = 2$
0	27	463666333	366636636	633336336
1	27	463663363	366633666	633333366
2	9	436666333	366663663	633363363
3	9	436636336	366633666	633333366
4	27	463666333	363633633	663366366
5	27	436666333	336633633	636366366
6	9	463666333	366366663	633363633
7	9	463636336	366336666	633333636

Table 1: The eight quotient multiplicity representatives.

union is the full 144-array set. The orbit sizes are $27, 27, 9, 9, 27, 27, 9, 9$; their stabilizers in the induced 27-element group have orders $1, 1, 3, 3, 1, 1, 3, 3$. For each array an explicit legal map sends it to its representative while preserving S_0. No search count is divided by one of these symmetry orders.

6.1 A centered coset invariant

For the centered fixed translate, let

$$m(v) = |D_0 \cap (v + H)| \qquad (v \in G/H \cong C_3^2).$$

We have $m(0) = 13$. The zero fiber for each of the four directions on G/H has count 40, so

$$m(v) + m(-v) = 27 \qquad (v \neq 0).$$

After singleton normalization, $m(v) = \sum_x r(x, v)$. The complete quotient classification gives $m(v) \in \{9, 12, 15, 18\}$ for $v \neq 0$. Legal normalizations induce an invertible linear map on G/H, so these statements transfer back to the centered fixed translate. Each opposite pair is consequently $\{9, 18\}$ or $\{12, 15\}$.

The $U = H$ moment identity is

$$\sum_v m(v)^2 = 121 + 60 \cdot 26 = 1681.$$

If j of the four opposite pairs are $\{9, 18\}$, then

$$1681 - 13^2 = j(9^2 + 18^2) + (4 - j)(12^2 + 15^2) = 1476 + 36j.$$

Thus $j = 1$: the eight off-H cosets have multiplicities

$$9, 18, 12, 12, 12, 15, 15, 15, \tag{9}$$

with one opposite pair $\{9, 18\}$ and three $\{12, 15\}$. All 144 arrays satisfy this invariant. It is a consequence of the classification, not an additional unexplained pruning condition. For an uncentered translate the distinguished thirteen-point coset can move; $|D \cap H| = 13$ is not asserted for an arbitrary translate of the original difference set.

7 Lifting completeness and the search partition

Fix a case in Table 1. Each of its 24 nonspecial K-cosets has nine points, partitioned into three scalar triples. A multiplicity-three cell must select exactly one triple, and a multiplicity-six cell exactly two. In either case there are exactly three choices, namely all members of the appropriate $\binom{3}{1}$ or $\binom{3}{2}$ family.

Number a cell's triples $0, 1, 2$ by their least encoded point. A digit j in a multiplicity-three cell selects triple j; in a multiplicity-six cell it omits triple j. Order the cells by placing the twelve six-cells first in increasing Q-index, followed by the twelve three-cells in increasing Q-index. Denote the resulting set for a case c and digit word $\boldsymbol{d} \in \{0,1,2\}^{24}$ by $\mathcal{L}(c, \boldsymbol{d})$.

Lemma 2. *For a fixed case, the map $\boldsymbol{d} \mapsto \mathcal{L}(c, \boldsymbol{d})$ is a bijection onto all scalar-fixed subsets with singleton part S_0 and that quotient multiplicity array. Every such set has 121 points.*

Proof. The union of S_0 and the chosen cell options has size $13 + 12 \cdot 6 + 12 \cdot 3 = 121$, is scalar-fixed, and has the prescribed multiplicities. Conversely, any fixed set with those multiplicities intersects each cell in an invariant subset of the required size, hence in one of the three stated options. This proves surjectivity. Intersecting the union with each of the disjoint cells recovers its option and unique digit, proving injectivity. □

There are $3^{24} = 282{,}429{,}536{,}481$ syntactic leaves per case before difference-count pruning. The bijection does not require enumeration of all these leaves. Different representative arrays give disjoint families because their quotient arrays differ; no assertion is made that all these leaves represent distinct affine-equivalence classes.

The local families have been checked independently of the search tables: scalar orbits were reconstructed from modular coordinates and all their required subfamilies formed, then compared as sets with every option emitted by each compiled search kernel. A separate completeness check also enumerated all three-point and six-point subsets of each nine-point cell and tested scalar closure directly. Both checks cover all $8 \cdot 24 \cdot 3 = 576$ labeled options and the thirteen initial points. The comparison definition comes from scalar fixedness and subset size, rather than from treating an existing option table as complete.

Split each 24-digit word after twelve digits. With leading zeroes retained, give its first half the big-endian rank

$$p = \sum_{j=0}^{11} d_j 3^{11-j}, \qquad \mathrm{id} = c\,3^{12} + p.$$

The complete ID domain has size

$$N = 8 \cdot 3^{12} = 4{,}251{,}528,$$

and is partitioned into 1,038 intervals

$$I_r = [4096r, \min(4096(r+1), N)), \qquad 0 \le r < 1038. \tag{10}$$

Each ID specifies the full cylinder of all 3^{12} possible tails. Different prefixes define disjoint cylinders and every full digit word has exactly one prefix, so these cylinders partition all eight trees. A prefix rejected before depth twelve must still have an explicit completed record; pruning is not a reason to omit its ID from the partition.

Separate checks reconstruct every prefix's rank, digits, and cylinder endpoints and compare them with both complete decoder output streams. They check all N IDs and exact endpoints,

not a sample or aggregate length alone. The interval walk checks that the first endpoint is zero, each start equals the previous stop, and the last stop is N. Those finite comparisons establish the contents of the supplied decoder outputs; identifying the outputs with executions of the intended compiled decoders is an external provenance assertion, not a Lean theorem.

8 Sound pruning and exhaustive computation

If $P \subseteq P'$, then $d_P(g) \le d_{P'}(g)$ for every g. Thus a partial set for which some nonzero difference count exceeds 60 has no difference-set extension. This is the only difference-count rejection needed by the search. At a complete 121-point leaf,

$$\sum_{g \neq 0} d_P(g) = 121 \cdot 120 = 242 \cdot 60.$$

Consequently all counts being at most 60 is equivalent to every count being exactly 60. The direct kernel also checks the exact leaf condition explicitly.

All partial sets are unions of scalar orbits, so $d_P(61g) = d_P(g)$. Reversing ordered pairs gives $d_P(-g) = d_P(g)$ as well. There are 36 classes outside H under $g \mapsto \pm 61^i g$, each of size six, and thirteen nonzero classes inside H, each of size two. Thus there are 49 nonzero difference classes. The compressed kernel stores the common count on each class and uses precomputed option contributions. The direct kernel instead adds and removes points and maintains the full elementwise difference vector. Compression is justified by constancy of contributions on these classes, rather than by averaging counts.

For every prefix, depth-first search tries all three digits at each remaining position, unless the monotone rejection rule applies. An explicit completion state is required for each prefix; an absent result, interruption, or partial continuation is not evidence of exhaustion. Production searches use the bound 60 and exclude the relaxed-bound features used for controls.

The two production passes agree on the complete ordered rows of ID, prefix-validity, node count, and solution count for all 4,251,528 IDs, and on node counts in every one of the 1,038 ranges. With their common node-count convention, the totals are

depth-twelve prefixes	4,251,528
search nodes	108,733,222,341
difference-set leaves	0.

These are traversal counts, not numbers of inequivalent subsets or an assertion that all $8 \cdot 3^{24}$ leaves were visited individually.

A complete additional direct replay reproduced every range. A separately written third lifting kernel also exhausted all 1,038 ranges, matching each range's validity, node, and solution counts and the digest of its canonical per-prefix row stream. That kernel has its own recursive traversal, difference arithmetic, option construction, ID decoder, and driver. Its six-cell labeling convention was selected by testing the two conventions against one range; this fixes labels only. Permuting the labels cannot add or remove a subset, and the complete pass then matched every range.

The compact computational certificate records each interval and its counts and row-stream digest. A static check verifies interval geometry and consistency, while replay recomputes the rows and compares their counts and digests. The digest binds a comparison to a particular row stream; it is not a proof of a universal nonexistence assertion. The exhaustive executions and their arithmetic and coverage checks supply the external computational part of the proof.

Proof of Theorem 1. Suppose D were such a difference set. Lemma 1 gives a fixed translate. The singleton conditions and complete classification send its H-part legally to S_0. The incidence inversion places its quotient in the complete 144-array set, and a legal S_0-stabilizing map sends that array to Table 1. By Lemma 2, the resulting difference set is a leaf of that case. Its prefix lies in exactly one interval (10). Exhaustion rejects every leaf under that interval by tests that reject no difference set, a contradiction. □

9 Completeness checks, controls, and independence

Agreement of the production kernels alone does not establish that their shared decomposition is complete. The proof uses the full local option-family comparisons, the separately written exhaustive lifting pass, the independent finite cover, and the Lean coverage implication, in addition to production agreement.

The finite cover checks exact equality of the singleton sets and legal-map orbit, the derived quotient set and covered case orbits, every local option family, all prefix cylinders, and the range partition. Three semantic gap controls exercise these obligations. One deletes ID 4095 and compensates by duplicating another ID while preserving the number of interval records and their summed length. A second duplicates one legal option while retaining its cell and cardinality. A third duplicates a case of the same orbit size. Each is rejected with a missing ID, subset, or quotient image witness, respectively. Their rejection does not rely on unchanged file digests or aggregate counts.

Further mutation tests reject altered scalar partitions, missing maps, nonimages in the case list, and two representatives from one quotient orbit. Replacing a representative by another member of its own orbit is accepted, as it should be. An altered range digest is rejected by both fresh production kernels. A changed node count, even with its aggregate total updated consistently, is rejected by fresh replay although it passes the static consistency check. These tests distinguish semantic checking from integrity and formatting checks.

There is a genuine positive discovery control at parameters $(121, 40, 13)$ in C_{121}: three generic implementations, without a supplied witness, recover 88 multiplier-invariant difference sets at 996,935 nodes. These are invariant sets without equivalence reduction. The same parameters in C_{11}^2 give a negative control; that negative outcome is not positive discovery evidence for the present search.

The actual order-243 ternary path has additional positive controls. Eight independently constructed fixed 121-point sets, one in each case, are supplied through forced digit paths with relaxed test-only bound $M = \max_{g \neq 0} d_P(g)$. Both production kernels recover their exact points at their specified depth-twelve IDs and reject bound $M - 1$:

c	id	M
0	91,615	66
1	815,056	72
2	1,486,729	69
3	1,750,181	72
4	2,473,622	69
5	3,145,295	69
6	3,214,101	69
7	3,937,542	72

For example, the case-zero word is

`011122200011122200011122`.

It has prefix ID 91,615 and $M = 66$. Each test returns one exact witness and thirteen tail nodes. These are bounded-count lifting and witness controls, not $(243, 121, 60)$ difference sets: their bounds are 66–72. Neighboring IDs are checked against their own reconstructed sets; a neighbor is not assumed to have no solution at a relaxed bound. Tests of omitted options, serialized continuation versus uninterrupted execution, and residual symmetries also agree. Residual symmetries are used for controls, not for reducing the production search domain.

A separate 10,000-ID comparison matches all per-ID rows between the production kernels and an additional coordinate-based implementation. This is a sample control and does not replace the complete third-kernel pass or prove coverage of a range.

The scope of independence is specific. The compressed and direct production kernels share a compilation unit, task and serialization code, prefix convention, S_0, scalar-partition and case inputs, cell order, and option contents. Their difference representations, updates, and traversals differ; they are not independent derivations of the entire reduction. The third complete kernel shares no search or certificate-driver code with them, but reads the same validated scalar-partition and case files and uses S_0. It reconstructs its own options and validates orbit membership; it does not itself discover the singleton class or quotient representatives.

Those common mathematical inputs are instead validated by separate programs that regenerate scalar orbits, enumerate all singleton candidates, classify quotient arrays in two different ways, and check legal full-group maps and exact case coverage. Thus the evidence composes independent input-completeness checks with a complete separate lifting exhaustion. A complete search fed directly by a newly self-generated chain of all front-end data is not claimed. The common mathematical implications are justified by the arguments and formalization, not inferred from matching search totals.

10 The exact formalization boundary

The Lean definition of the group is the product of the three residue groups of moduli $3, 9, 9$. Its `IsDiffSet` predicate is exactly cardinality 121 and ordered-difference count 60 at every nonzero element. The formalization proves the first-multiplier specialization by the classical group-ring argument, as well as centering, the index-three moments, the singleton congruences, legal normalizations, quotient inversion, lifting, pruning, and ID partition.

In particular, `coverage` proves the following implication, with no hypotheses about outputs of a finite enumerator or of a search: if D is a difference set, then there exist $u \in G$, a word w of legal affine maps, an integer $r < 1038$, an $\mathrm{id} \in I_r$, and a twelve-digit tail $\boldsymbol{t}$ such that

$$F_w(D - u) = \mathcal{L}\left(\left\lfloor \frac{\mathrm{id}}{531441} \right\rfloor, \mathrm{digits}(\mathrm{id}, \boldsymbol{t})\right). \tag{11}$$

Here the first twelve digits are the leading-zero-preserving base-three expansion of $\mathrm{id} \bmod 531441$ and the last twelve are $\boldsymbol{t}$. The map F_w is an additive automorphism followed by a translation in H. This is an existence statement; the normalizing map or normalized representative for the original D need not be unique. The interval of each ID is unique.

The formal singleton proof uses its own finite classification: fourteen layer-zero representatives, 64 layer-one shear classes, 52,220 patterns, and twelve survivors with legal words to S_0. These numbers describe that classification, not the 5,335,092-mask enumeration in (4). The quotient proof checks all 2^{13} assignments and the legal witnesses to the eight actual case rows. Generated witness tables have their required properties and completeness checked in the kernel; they are not imported as axioms.

For an interval I_r, define the semantic assertion

$$\begin{aligned}\mathrm{RangeClear}(r): \quad & \text{for every } \mathrm{id} \in I_r \text{ and every twelve-digit tail } \boldsymbol{t}, \\ & \mathcal{L}\left(\left\lfloor \frac{\mathrm{id}}{531441} \right\rfloor, \mathrm{digits}(\mathrm{id}, \boldsymbol{t})\right) \text{ is not a difference set.}\end{aligned} \tag{12}$$

The final Lean theorem `no_difference_set` is conditional on the family

$$\forall r < 1038, \qquad \mathrm{RangeClear}(r).$$

These are 1,038 semantic assertions. A receipt formulation packages each assertion as the `clear` field of $\mathrm{Receipt}(r, \mathrm{digest}_r)$; the digest is only a label, not additional mathematical evidence. Lean does not compute SHA-256 here, interpret C++ output, or derive the assertions from digest strings. The external exhaustive computations in Section 8 supply that part of the computer-assisted proof.

Equality of the Lean leaf table with the implemented search options is also an external comparison, covering all 576 options and thirteen singleton points, strengthened by the complete compiled option dumps and separate definition-based checks. It is neither an omitted hypothesis of `coverage` nor a formal proof of the C++ implementation, compiler, or execution environment. The finite cover's decoder comparisons likewise retain the execution-provenance boundary stated in Section 7.

The recorded kernel checks of the reductions and conditional final theorem use only `propext`, `Classical.choice`, and `Quot.sound`. Planted formal errors are rejected: dropping the last range, pruning when a count reaches λ rather than exceeds it, weakening the exact leaf criterion to $\lambda + 1$, and omitting an admissible singleton survivor. The first three false statements also have formal counterexamples. These checks establish the stated reduction scope; they do not turn the external search into a whole-theorem Lean proof from the definitions.

References

[1] K. T. Arasu and S. L. Ma, *A nonexistence result on difference sets, partial difference sets and divisible difference sets*, Journal of Statistical Planning and Inference **95** (2001), 67–73. doi:10.1016/S0378-3758(00)00278-0.

[2] Y. Q. Chen, Q. Xiang, and S. K. Sehgal, *An exponent bound on skew Hadamard abelian difference sets*, Designs, Codes and Cryptography **4** (1994), 313–317. doi:10.1007/BF01388647.

[3] Daniel M. Gordon, *The La Jolla Difference Set Repository*, presentation, August 3, 2019, slides 11 and 38 (slide 38 is physical PDF page 49). https://cargo.wlu.ca/ArasuFest/ArasuFest_talks/Dan_Gordon.pdf.

[4] Daniel M. Gordon, *La Jolla Difference Sets Repository*, archived dataset, version 1.3, April 24, 2026. doi:10.5281/zenodo.19740740. Repository and definition of entries: https://dmgordon.org/difference-sets/.

[5] Marshall Hall Jr., *A Survey of Difference Sets*, Proceedings of the American Mathematical Society **7** (1956), 975–986. doi:10.1090/S0002-9939-1956-0082502-7.

[6] Zhiwen He and Jiayan Wu, *Nonexistence results and constructions for signed difference sets*, preprint, arXiv:2608.22936v2, September 20, 2026 (first version August 24, 2026). https://arxiv.org/abs/2608.22936v2.

[7] E. S. Lander, *Symmetric Designs: An Algebraic Approach*, London Mathematical Society Lecture Note Series, vol. 74, Cambridge University Press, Cambridge, 1983. doi:10.1017/CBO9780511662164.

[8] Emma Lehmer, *On residue difference sets*, Canadian Journal of Mathematics **5** (1953), 425–432. doi:10.4153/CJM-1953-047-3.

[9] K. H. Leung, S. L. Ma, and B. Schmidt, *Nonexistence of abelian difference sets: Lander's conjecture for prime power orders*, Transactions of the American Mathematical Society **356** (2004), 4343–4358. doi:10.1090/S0002-9947-03-03365-8.

[10] K. H. Leung and B. Schmidt, *The Field Descent Method*, Designs, Codes and Cryptography **36** (2005), 171–188. doi:10.1007/s10623-004-1703-7.

[11] Niklas Miller, *Forbidden multipliers in abelian difference sets*, preprint, arXiv:2511.10231v1, November 13, 2025. `https://arxiv.org/abs/2511.10231v1`.

[12] R. E. A. C. Paley, *On Orthogonal Matrices*, Journal of Mathematics and Physics **12** (1933), 311–320. doi:10.1002/sapm1933121311.

[13] R. J. Turyn, *Character sums and difference sets*, Pacific Journal of Mathematics **15** (1965), 319–346. doi:10.2140/pjm.1965.15.319.

[14] A. Vera López and M. A. García Sánchez, *On the existence of abelian difference sets with* $100 < k \leq 150$, Journal of Combinatorial Mathematics and Combinatorial Computing **23** (1997), 97–112; Table 1, p. 105. Publisher's article record.